From Ideas to Action

Three agents, three problems

October 2026

Motivation: producing anything takes three kinds of work

A **strategist** decides *what* to do.

A **solver** works out *how*.

An **executor** *does* it.

> Which of the three can AI improve, and does that make the economy grow?

This paper

- **Model.** Production is a pipeline of three agents: a strategist who screens plans (Napoleon), a solver who approximates and optimizes (Berthier), and an executor bounded by physical technology (Lannes).
- **Progress.** Each agent can improve: AI improves Berthier, vision improves Napoleon, and physical progress improves Lannes.
- **Results.** Bounds on the level and the growth rate of output under each kind of progress, alone and combined.
- **Applications.** Production, where everything must be built, and research, where nothing has to be; with a numerical illustration.

Main results

1. AI alone raises the level of output to a ceiling, not its growth rate.
2. Vision has a ceiling too: with thin-tailed plan values and Lannes fixed, even RSI levels off.
3. The physical frontier bounds output, and ambition beyond it is wasted.
4. AI and physical progress together lift output until Napoleon becomes the bottleneck.
5. In **research**, where nothing has to be built, AI and vision together sustain growth.

In the illustration, over 60 years: AI multiplies output by 2.8, AI with physical progress by 9, and all three improvements by more than 6,000.

Related literature

- **Ideas and growth.** Romer (1990); Jones (1995); Weitzman (1998).
- **Research as search.** Kortum (1997): sustained growth from search needs Pareto-tailed ideas, as in our Result 3.
- **AI and growth.** Aghion, Jones and Jones (2019) on bottlenecks; Acemoglu and Restrepo (2018) on tasks.
- **Knowledge in organizations.** Garicano (2000): hierarchies that separate problem solving from production.
- **Sequential information acquisition.** Wald (1947): when to stop investigating a plan.

> We separate judgment, solving, and execution.

Outline

1. Introduction
2. Model
3. Results
4. Application: production
5. Application: research
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Plans, actions, and value

A **plan** ξ says what to do. A detailed **action** $a \in \mathcal{A}(\xi)$ carries it out and yields **payoff** $W(a;\xi)$.

$$a^*(\xi) \in \arg\max_{a \in \mathcal{A}(\xi)} W(a;\xi), \qquad v(\xi) = W\big(a^*(\xi);\xi\big)$$

> $a^*(\xi)$ is the **ideal solution**; $v(\xi)$ is the **value** of the plan.

Throughout, every problem is smooth: $\mathcal{A}(\xi) \subset \mathbb{R}^d$ is compact and $W(\cdot\,;\xi)$ is infinitely differentiable (C^∞) in a.

Three agents, three problems

	Problem	Input	Output
Strategist	What should we do?	candidate plans	plan ξ
Solver	Exactly how is ξ done?	plan ξ	solution $\hat{a}$
Executor	Carry out $\hat{a}$ physically	solution $\hat{a}$	action $\tilde{a}$

> Output is $Y = W(\tilde{a}; \xi)$: the payoff of what actually gets done.

Meet the three agents

Napoleon the strategist
decided what each campaign should achieve.

Berthier the solver
his chief of staff: turned Napoleon's intentions into detailed orders, routes, and timetables.

Lannes the executor
the marshal who carried those orders out on the battlefield.

From here on we call the three agents by their names.

Where candidate plans come from

Candidate plans are drawn independently, $\xi \sim F$. Each comes with its *whole* payoff function $W(\cdot\,; \xi)$.

A plan's **expected value** is computed from that function and from how it will be solved and carried out:

$$U = u(\xi) = \mathbb{E}\big[W(\tilde{a}; \xi) \mid \xi\big].$$

> G, the distribution of U, is *induced*: by the pool of plans F, and by how well Berthier and Lannes handle each plan.

In general G can be any distribution. It is normal only under the conditions of the Gaussian example in the appendix.

The strategist investigates before choosing

Definition 1 (Strategist, or Napoleon)

A strategist, or a Napoleon, is a pair $N = (\tau_s, \mathcal{S})$:

- **signal technology** $\mathcal{S} = \{P(\cdot \mid u)\}_u$: signals $s_1, s_2, \cdots \in \Sigma$ about a plan's expected value $U \sim G$, i.i.d. given U;
- **signal time** τ_s: how long each signal takes.

> Plans are plentiful. Time to learn which are good is scarce.

Napoleon judges what a plan will actually deliver, after the losses in solving and doing it, not its ideal value.

A policy says when to stop investigating

A **policy** is a decision after every history of signals:

$$\pi = (\pi_n)_{n \geq 1}, \qquad \pi_n : \Sigma^n \to \{\text{accept}, \text{ reject}, \text{ wait}\}.$$

It stops after $N_\pi = \min\{n : \pi_n(s_1, \dots, s_n) \neq \text{wait}\}$ signals.

Π is the set of policies with $\mathbb{E}[N_\pi] < \infty$.

The solver works out the details

Definition 2 (Solver, or Berthier)

A solver, or a Berthier, is a pair $B = (\tau_B, k)$. Berthier replaces the true payoff with its best approximation from a **sieve** $\mathcal{W}_k$, a family of functions of complexity k:

$$\hat{W}(\cdot\,; \xi) \in \arg\min_{f \in \mathcal{W}_k} \sup_{a \in \mathcal{A}(\xi)} |f(a) - W(a; \xi)|.$$

- **sieve size** k: how rich a model Berthier can handle;
- **solving time** τ_B per plan, so plans are solved at rate $\mu_B = n_B/\tau_B$.

A richer sieve is a better approximation

The **approximation error** is the distance from W to the sieve:

$$\varepsilon_B(k) \equiv \min_{f \in \mathcal{W}_k} \sup_{a \in \mathcal{A}(\xi)} |f(a) - W(a;\xi)|.$$

For standard sieves (polynomials, splines) and $W \in C^s$, approximation theory gives

$$\varepsilon_B(k) \;\leq\; C_s\, k^{-s/d} \qquad \text{for every } s, \text{ since } W \in C^\infty.$$

> The approximation is global: it does not depend on knowing where the optimum is.

C_s depends on bounds on the first s derivatives of W. Neural networks are sieves too. The rate worsens with the number of decision variables d.

Berthier's problem

Problem B (approximate, then solve)

Solve the approximate problem exactly:

$$\hat{\mathcal{P}}(\xi): \quad \hat{a} \in \arg\max_{a \in \mathcal{A}(\xi)} \hat{W}(a;\xi).$$

Solution loss: what the approximation costs in the true problem,

$$\kappa_B(\xi) = \left|v(\xi) - W(\hat{a};\xi)\right| = v(\xi) - W(\hat{a};\xi).$$

The two agree because $v(\xi)$ is the maximum, so $W(\hat{a};\xi) \le v(\xi)$.

> Berthier errs in his model of the problem, never in the goal.

A good approximation gives a good solution

Lemma 1 (approximation bound)

If Berthier's approximation is within $\varepsilon_B(k)$ of the true payoff, then

$$\kappa_B(\xi) \;\le\; 2\,\varepsilon_B(k) \;\le\; 2\,C_s\,k^{-s/d}.$$

> Solution loss is at most twice the approximation error, whatever the problem.

The executor carries out the solution

Definition 3 (Executor, or Lannes)

An executor, or a Lannes, is a pair $L = (\tau_L, \mathcal{R})$:

- **budget set** $\mathcal{R} \subseteq \mathcal{A}(\xi)$: the actions that can physically be carried out;
- **execution time** τ_L per plan, so plans are carried out at rate $\mu_L = n_L/\tau_L$.

> Berthier's solution $\hat{a}$ need not lie in $\mathcal{R}$.

Lannes's problem

Problem L (execution)

Carry out the feasible action closest to Berthier's solution:

$$\tilde{a} = \arg\min_{r \in \mathcal{R}} \|r - \hat{a}\|.$$

Execution loss: zero whenever $\hat{a} \in \mathcal{R}$,

$$\kappa_L(\xi) = \mathbb{E}\big[W(\hat{a};\xi) - W(\tilde{a};\xi) \mid \xi\big].$$

Best feasible value: $\bar{U}(\xi) = \max_{r \in \mathcal{R}} W(r;\xi) \le v(\xi)$.

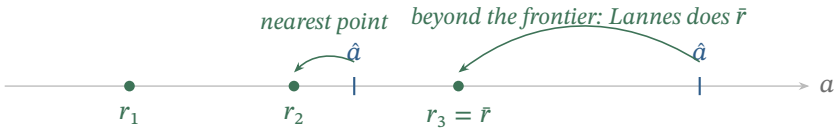
Example: a one-dimensional budget set

Let actions be real numbers, $\mathcal{A} = [\underline{a}, \overline{a}]$. Berthier can propose any $\hat{a} \in \mathcal{A}$, but Lannes's budget set is finite:

$$\mathcal{R} = \{r_1 < r_2 < \cdots < r_J\}, \qquad \tilde{a} = \arg\min_{r \in \mathcal{R}} |r - \hat{a}|.$$

> $\mathcal{R}$ is the set of physical technologies that exist.

The physical frontier



> The **physical frontier** is $\bar{r} = \max \mathcal{R}$. Anything beyond it is done at $\bar{r}$.

Output depends on all three



$$U = \mathbb{E}[Y \mid \xi] = v(\xi) - \kappa_B(\xi) - \kappa_L(\xi)$$

> Expected value: ideal value, minus what is lost solving it, minus what is lost doing it.

A local view of the losses

Near the ideal solution, with $d = a - a^*(\xi)$:

$$W(a;\xi) \approx v(\xi) - \frac{1}{2}\, d^\top H\, d, \qquad H = -\nabla_a^2 W\big(a^*(\xi);\xi\big).$$

If Berthier's approximation puts the optimum at $\hat{a} = a^*(\xi) + \delta_B$, his loss is $\kappa_B = \frac{1}{2}\delta_B^\top H \delta_B$. Lannes's loss is the cost of moving $\hat{a}$ into $\mathcal{R}$.

> Berthier's loss comes from approximation; Lannes's loss comes from infeasibility.

Flows and capacity

Accepted flow: plans accepted per unit of time by n_A Napoleons,

$$Q(\pi) = \frac{n_A \, P_\pi(\text{accept})}{\tau_s \, \mathbb{E}_\pi[N_\pi]}.$$

Downstream capacity: plans per unit of time that n_B Berthiers and n_L Lannes can handle,

$$K = \min\left\{\frac{n_B}{\tau_B}, \; \frac{n_L}{\tau_L}\right\}.$$

$Q(\pi)$ is a rate: n_A/τ_s signals per unit of time, divided by $\mathbb{E}_\pi[N_\pi]$ signals per plan, times the share accepted. Derivation in the appendix.

Napoleon's problem

Problem N (investigation)

$$\max_{\pi \in \Pi} \; Q(\pi)\, \mathbb{E}_\pi[U \mid \text{accept}] \qquad \text{s.t.} \qquad Q(\pi) \leq K$$

By iterated expectations, $\mathbb{E}_\pi[Y \mid \text{accept}] = \mathbb{E}_\pi[U \mid \text{accept}]$.

> Waiting buys information about one plan, but leaves less time for others.

Four kinds of progress

	Improves	Definition	Effect
AI	Berthier	$\tau_B' \leq \tau_B$ or $k' \geq k$	faster solving, richer approximations
Vision	Napoleon	$\tau_s' \leq \tau_s$ or $\mathcal{S}'$ Blackwell more informative	faster screening, sharper judgment
RSI	Napoleon and Berthier	AI and vision together	both of the above
Physical progress	Lannes	$\mathcal{R}' \supseteq \mathcal{R}$ or $\tau_L' \leq \tau_L$	frontier moves out, capacity rises

> Only physical progress changes Lannes and moves the frontier.

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Assumptions behind the results

1. For Results 1–3, Lannes is fixed: τ_L and $\mathcal{R}$ do not change.
2. $\mathbb{E}|v(\xi)| < \infty$, and output starts positive.
3. For Results 2–3: best feasible values $\max_{r \in \mathcal{R}} W(r; \xi)$ have thin tails (σ-sub-Gaussian around their mean); and screening outpaces capacity, $n_A/\tau_s \geq \sqrt{e}\, n_L/\tau_L$.

> Whatever Berthier does, the action lies in $\mathcal{R}$, so $U \leq \max_{r \in \mathcal{R}} W(r; \xi)$.

Output growth rate: $g_y = \limsup_t \frac{1}{t} \ln y_t$. Proofs are in the appendix.

Result 1: AI alone cannot sustain growth

Proposition 1

Fix Napoleon and Lannes. Along any path of AI, for all t,

$$y_t \;\le\; \frac{n_A}{\tau_s}\,\mathbb{E}_{\xi\sim F}\Big[\max\Big\{0,\ \max_{r\in\mathcal{R}} W(r;\xi)\Big\}\Big] \;<\; \infty.$$

> AI raises output at most to a fixed ceiling, then stops.

n_A/τ_s: the fastest rate of examining plans. $\max_{r\in\mathcal{R}} W(r;\xi)$: the plan's best feasible value. Neither depends on Berthier.

No information beats perfect sorting

Lemma 2 (selection bound)

Let $Z - \mathbb{E}[Z]$ be σ-sub-Gaussian, and let A be any event with $P(A) = q$, such as "Napoleon accepts the plan". Then

$$\mathbb{E}[Z \mid A] \;\le\; \mathbb{E}[Z] + \sigma\sqrt{2\ln(1/q)}.$$

> Being more selective helps, but only like $\sqrt{\ln(1/q)}$.

Applied with $Z = \max_{r \in \mathcal{R}} W(r; \xi)$, it bounds the plans Napoleon accepts under any signals and any policy.

Result 2: sharper judgment has a ceiling

Proposition 2

Fix the signal time τ_s. For any Berthier and any signal technology, even perfect information,

$$\mathcal{Y} \;\le\; K\left(\max\left\{0,\; \mathbb{E}_{\xi\sim F}\left[\max_{r\in\mathcal{R}} W(r;\xi)\right]\right\} + \sigma\sqrt{2\ln\frac{n_A}{K\tau_s}}\right).$$

> Better judgment can at best reach full information.

Result 3: thin tails stop all growth

Proposition 3

If signals speed up at most exponentially, $\tau_{s,t} \geq \tau_{s,0} e^{-gt}$, then for any AI and any vision

$$y_t \;\leq\; \frac{n_L}{\tau_L}\left(\max\left\{0,\; \mathbb{E}_{\xi\sim F}\left[\max_{r\in\mathcal{R}} W(r;\xi)\right]\right\} + \sigma\sqrt{2\ln\frac{n_A\tau_L}{n_L\tau_{s,0}} + 2gt}\right).$$

> Output grows at most like $\sqrt{t}$: AI, vision, and RSI all level off.

Heavy tails change this

Suppose expected values have a Pareto tail, $P(U > x) = (x/x_m)^{-\alpha}$ with $\alpha > 1$. Then the best plans improve as a power of the pool screened:

$$M(q) = \frac{\alpha}{\alpha - 1} x_m \, q^{-1/\alpha}, \qquad g_y \;\leq\; g/\alpha,$$

with equality under perfectly informative signals.

> Lasting growth from vision needs faster screening *and* rare, enormously valuable plans.

Better plans sit further out

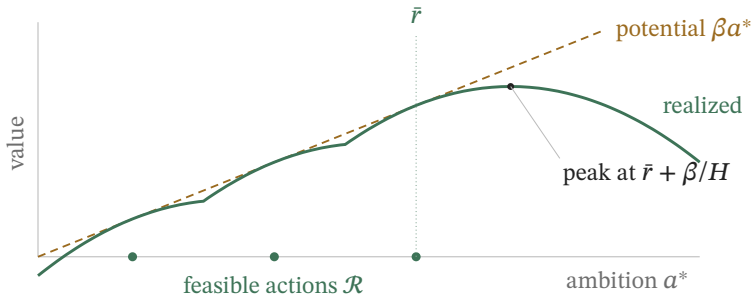
A plan's ideal action measures its **ambition**, and value rises with ambition:

$$v(\xi) = \beta\, a^*(\xi), \qquad \beta > 0.$$

The payoff is as before:

$$W(a;\xi) = v(\xi) - \frac{H}{2}\big(a - a^*(\xi)\big)^2, \qquad H > 0.$$

Ambition beyond the frontier is wasted



Napoleon's signals are about the realized curve, U, not the potential line, so he does not chase plans beyond the frontier.

Result 4: the physical frontier bounds output

Proposition 4

For any Napoleon, any Berthier, and any distribution of plans,

$$U,\ Y \ \le\ \beta\bar{r} + \frac{\beta^2}{2H} \qquad \text{and} \qquad y \ \le\ \mu_L\Big(\beta\bar{r} + \frac{\beta^2}{2H}\Big).$$

> Even heavy-tailed plans cannot help: their tail lies beyond the frontier.

Ambition and the frontier are complements

For a plan beyond the frontier, $a^* > \bar{r}$, Lannes does $\bar{r}$ and the realized value is

$$Y(a^*, \bar{r}) = \beta a^* - \frac{H}{2}(a^* - \bar{r})^2.$$

It has **increasing differences** in ambition and the frontier:

$$\frac{\partial^2 Y}{\partial a^* \, \partial \bar{r}} = H > 0, \qquad \arg\max_{a^*} Y = \bar{r} + \frac{\beta}{H}.$$

> Moving the frontier is worth most for the most ambitious plans, and the best plans sit just beyond it.

In the Gaussian example (appendix), $W = v(\xi) + w(a)$ and the cross-partial is zero: there physical progress works only through κ_L and capacity.

Without physical progress, nothing grows

Corollary

If $\bar{r}$ and μ_L grow at most at rates g_R and g_L, then for any AI and any vision

$$\limsup_{t\to\infty} \frac{1}{t} \ln y_t \;\le\; g_R + g_L.$$

> No physical progress means no long-run growth, under AI, vision, or RSI.

Result 5: then Napoleon becomes the ceiling

Proposition 5

Fix Napoleon. Along any path of AI and physical progress, for all t,

$$y_t \;\le\; \frac{n_A}{\tau_s}\,\mathbb{E}_{\xi\sim F}\big[\max\{0,\; v(\xi)\}\big],$$

and once $\min\{\mu_{B,t}, \mu_{L,t}\} \ge n_A/\tau_s$*, Napoleon's capacity constraint is slack.*

> Together they make Napoleon the bottleneck: a higher level, but no lasting growth.

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Parameters

	Year 0	Each year after
Napoleon	$m = 0, \sigma = 4, r = 4, \tau_s = 0.05$	vision: τ_s, r divided by 1.1
Berthier	$\mu_B = 1, \kappa_B = 1.5$	AI: $\mu_B \times 1.25$, $\kappa_B \div 1.25$ (rising k)
Lannes	$\mu_L = 2, \kappa_L = 1$	unchanged

One Napoleon ($n_A = 1$). Gaussian plans (appendix), so both losses are constants and $U \sim N(-\kappa, 4^2)$. Napoleon re-chooses the number of signals each year. Illustrative parameters, not a forecast.

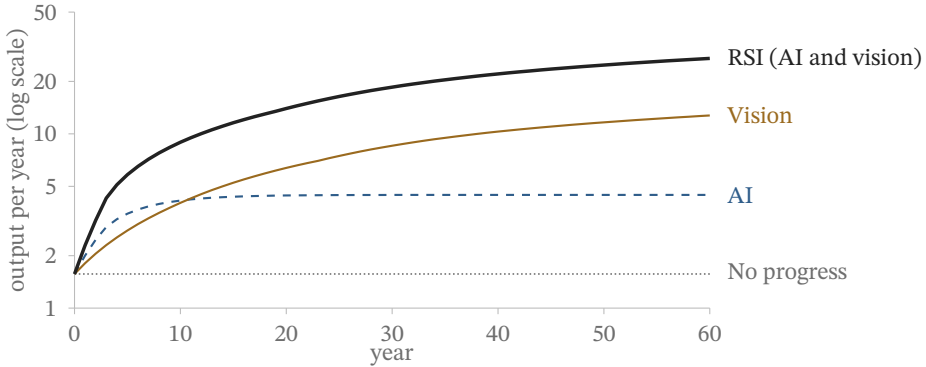
Year 0: how many signals Napoleon takes

n signals	R_n examined	q_n share accepted	$\sqrt{\rho_n}$ accuracy	$S(q_n)$ selectivity	$y(n)$ output
1	20.0	0.05	0.45	2.06	1.19
2	10.0	0.10	0.58	1.76	1.55
3	**6.7**	**0.15**	**0.65**	**1.55**	**1.57**
4	5.0	0.20	0.71	1.40	1.46*

> Napoleon takes 3 signals: with fewer he cannot tell plans apart; with more he sees too few plans.

$m_U = -2.5$, $\sigma_U = 4$, $r = 4$, $\tau_s = 0.05$, $n_A = K = 1$. *Cutoff hits zero: fewer than K accepted. Derivation in the appendix.

With thin tails, every path levels off



What the illustration shows

	Year 0	10	20	60	Growth, years 50–60
No progress	1.57	1.57	1.57	1.57	0%
AI	1.57	4.14	4.43	4.47	0%
Vision	1.57	4.02	6.38	12.75	0.9%
RSI	1.57	8.95	13.98	27.12	0.9%

> AI hits its ceiling. Vision and RSI keep rising, but ever more slowly.

Better Berthier and better Lannes, Napoleon fixed

	Year 0	10	20	60	Binds at year 60
No progress	1.57	1.57	1.57	1.57	Berthier
AI	1.57	4.14	4.43	4.47	Lannes
Physical progress	1.57	2.46	2.56	2.57	Berthier
AI + physical	1.57	11.75	13.99	**14.27**	Napoleon
All three, with vision	1.57	31.59	168.1	9,654	none: grows 10.1%

> Alone, AI adds 2.9 and physical progress 1.0. Together they add 12.7, then level off until vision joins.

Illustration parameters, plus $\mu_L \times 1.25$ and $\kappa_L \div 1.25$ a year.

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A research sector: Napoleon and Berthier only

In **research** nothing has to be built: Napoleon picks the question and Berthier's solution *is* the output.

$$Y = W(\hat{a};\xi), \qquad U = v(\xi) - \kappa_B(\xi), \qquad K = \mu_B = \frac{n_B}{\tau_B}.$$

> No budget set and no Lannes capacity: the only constraints are screening and solving.

Proofs, designs, software, simulations. Same model, with $\mathcal{R} = \mathcal{A}$ and $\mu_L = \infty$, so the best feasible value is $v(\xi)$ itself.

In research, AI relaxes Napoleon's constraint

Napoleon's capacity constraint is $Q(\pi) \leq K$.

- *Production*: $K = \min\{\mu_B, \mu_L\} \leq \mu_L$. AI can lift K only up to Lannes's capacity.
- *Research*: $K = \mu_B$. AI lifts K without limit.

Since $Q(\pi) \leq n_A/\tau_s$ for every policy, once $\mu_B \geq n_A/\tau_s$ the constraint never binds.

> Napoleon stops rationing: he pursues every plan he expects to be worth something.

Result 6: AI lifts research to Napoleon's ceiling

Proposition 6

In research, fix Napoleon. Along any path of AI, for all t,

$$y_t \;\le\; \frac{n_A}{\tau_s}\,\mathbb{E}_{\xi\sim F}\big[\max\{0,\; v(\xi)\}\big],$$

and once $\mu_{B,t} \ge n_A/\tau_s$, Napoleon's capacity constraint is slack.

> AI makes Napoleon the bottleneck. The ceiling depends only on how fast he can screen.

Result 7: in research, RSI sustains growth

Proposition 7

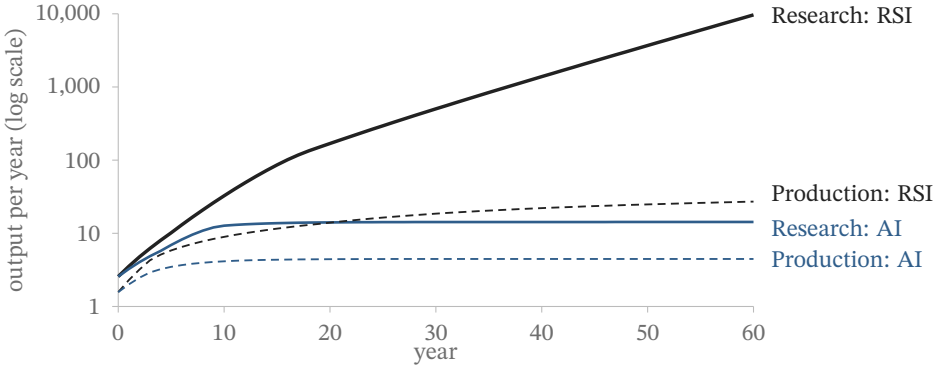
In research with Gaussian plans, suppose AI raises capacity at rate g_B, $\mu_{B,t} = \mu_{B,0}e^{g_B t}$, and vision speeds up screening at rate g, $\tau_{s,t} = \tau_{s,0}e^{-gt}$, without making signals noisier. Then

$$g_y = \min\{g_B,\ g\}.$$

> Research grows when *both* thinking constraints keep relaxing, even with thin tails.

AI alone ($g = 0$) or vision alone ($g_B = 0$) still gives zero growth: one constraint stays fixed.

Research versus production



Same parameters as the illustration; research drops Lannes ($\kappa_L = 0$, no capacity cap).

What changes in research

	Production		Research	
	Year 60	Growth	Year 60	Growth
AI	4.47	0%	14.27	0%
Vision	12.75	0.9%	13.75	0.8%
RSI	27.12	0.9%	9,654	10.1%

- With AI, Napoleon goes from accepting 1 plan a year to 10, and from 3 signals per plan to 1.
- With RSI, research output grows at the screening rate, about 10% a year.

Growth over years 50–60. Year 0 output is 1.57 in production and 2.57 in research.

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Summary

AI raises the level of output, not its growth rate.

Sharper judgment has a ceiling; faster screening sustains growth only if plan values are heavy-tailed.

The physical frontier bounds output, whatever thinking does.

In production, lasting growth needs physical progress, and AI with physical progress alone stops at Napoleon's ceiling.

In research, AI relaxes Napoleon's constraint, and RSI can sustain growth.

Next steps

1. **Endogenous complexity.** Do ambitious plans also need more Lannes time, $\tau_L(\xi)$, or more Berthier time, $\tau_B(\xi)$?
2. **Measurement.** Estimate the tail of plan values and the pace of physical progress; separate failures of judgment, solving, and execution in data.
3. **Occupational choice and wages.** How does cheaper Berthier labour change demand for Napoleon- and Lannes-like workers?
4. **General equilibrium.** Let the numbers of each agent, n_A, n_B, n_L, respond to prices.

> Does AI's first-order effect come from better ideas, or from realizing the existing stock of ideas?

Appendix

Computing Q: the acceptance probability

Write $s_{1:n} = (s_1, \dots, s_n)$ and $P_u^n(ds_{1:n}) = \prod_{k=1}^{n} P(ds_k \mid u)$. Napoleon accepts at signal n on

$$A_n = \{s_{1:n} : \ \pi_j(s_{1:j}) = \text{wait for all } j < n, \ \ \pi_n(s_{1:n}) = \text{accept}\}.$$

These events are disjoint, so

$$P_\pi(\text{accept} \mid u) = \sum_{n \geq 1} P_u^n(A_n), \qquad P_\pi(\text{accept}) = \int P_\pi(\text{accept} \mid u)\, dG(u).$$

$P_\pi(\text{accept} \mid u)$ is the chance a plan of value u is accepted; sharper signals make it rise more steeply in u.

Computing Q: the length of an investigation

Napoleon is still waiting after m signals on

$$W_m = \{s_{1:m} \ : \ \pi_j(s_{1:j}) = \text{wait for all } j \le m\}, \qquad W_0 = \text{everything}.$$

He takes an n-th signal exactly when he waited through the first $n-1$, so

$$\mathbb{E}_\pi[N_\pi] = \sum_{n\ge 1} P_\pi(N_\pi \ge n) = \sum_{n\ge 1} \int P_u^{n-1}(W_{n-1})\, dG(u).$$

One investigation takes $\tau_s N_\pi$ units of time, $\tau_s \mathbb{E}_\pi[N_\pi]$ on average.

Computing Q: putting it together

Each Napoleon examines independent plans one after another. By the renewal–reward theorem,

$$\lim_{T\to\infty} \frac{\#\{\text{plans examined by } T\}}{T} = \frac{1}{\tau_s\, \mathbb{E}_\pi[N_\pi]}.$$

A share $P_\pi(\text{accept})$ is accepted, and n_A Napoleons work in parallel:

$$Q(\pi) = \underbrace{n_A}_{\text{Napoleons}} \times \underbrace{\frac{1}{\tau_s\, \mathbb{E}_\pi[N_\pi]}}_{\text{plans examined each}} \times \underbrace{P_\pi(\text{accept})}_{\text{share accepted}}.$$

> Waiting longer raises $\mathbb{E}_\pi[N_\pi]$: better information, but fewer plans examined.

Computing Q: the value of accepted plans

The same objects give the objective in Problem N:

$$\mathbb{E}_\pi[U \mid \text{accept}] = \frac{\int u\, P_\pi(\text{accept} \mid u)\, dG(u)}{P_\pi(\text{accept})},$$

so output is

$$\mathcal{Y}(\pi) = Q(\pi)\, \mathbb{E}_\pi[U \mid \text{accept}] = \frac{n_A}{\tau_s\, \mathbb{E}_\pi[N_\pi]} \int u\, P_\pi(\text{accept} \mid u)\, dG(u).$$

The renewal–reward theorem

Suppose a process splits into independent cycles: cycle i lasts T_i and earns R_i, with (T_i, R_i) i.i.d. and $\mathbb{E}[T] < \infty$. Then almost surely

$$\lim_{t\to\infty} \frac{\text{reward earned by } t}{t} = \frac{\mathbb{E}[R]}{\mathbb{E}[T]}.$$

Why. By the law of large numbers, about $t/\mathbb{E}[T]$ cycles finish by time t, and their average reward tends to $\mathbb{E}[R]$.

Applied to Napoleon

One cycle is one investigation, $T_i = \tau_s N_i$. Reward 1 counts plans examined; reward $\mathbf{1}\{\text{accept}\}$ counts plans accepted; reward $U_i\mathbf{1}\{\text{accept}\}$ gives output.

Example: Gaussian plans

Suppose plans differ only in their *level*: a common shape w, with $\max_a w(a) = 0$, and a normal ideal value,

$$W(a;\xi) = v(\xi) + w(a), \qquad v(\xi) \sim N(m, \sigma^2).$$

Shifting W by a constant moves neither Berthier's sieve solution $\hat{a}$ nor Lannes's action $\tilde{a}$, so both losses are the same for every plan:

$$U = v(\xi) - \kappa_B - \kappa_L \;\sim\; N\big(m - \kappa,\ \sigma^2\big).$$

This needs the sieve to contain constant functions, as polynomials and splines do.

Example: Gaussian signals

Under the Gaussian-plans conditions, $U \sim N(m - \kappa, \sigma^2)$. Signals are normal around it:

$$s_k = U + \eta_k, \qquad \eta_k \sim N(0, \sigma_\eta^2).$$

After n signals, Napoleon's estimate of U is the posterior mean

$$\mu_n \sim N\big(m - \kappa,\ \rho_n \sigma^2\big), \qquad \rho_n = \frac{n}{n + r}, \qquad r = \frac{\sigma_\eta^2}{\sigma^2}.$$

ρ_n is the *accuracy* of the estimate. Below, $m_U = m - \kappa$ and $\sigma_U = \sigma$.

Q in the Gaussian case

Take n signals, then accept if the posterior mean clears a cutoff, $\mu_n \geq z$ (chosen on the next slide). The estimate is

$$\mu_n = m_U + \rho_n(\bar{s}_n - m_U), \qquad \bar{s}_n = \frac{1}{n}\sum_{k\leq n} s_k \sim N\big(m_U,\ \sigma_U^2 + \sigma_\eta^2/n\big),$$

so $\mathrm{Var}(\mu_n) = \rho_n^2(\sigma_U^2 + \sigma_\eta^2/n) = \rho_n\sigma_U^2$. With $N_\pi = n$,

$$P_\pi(\text{accept}) = 1 - \Phi\Big(\frac{z - m_U}{\sigma_U\sqrt{\rho_n}}\Big), \qquad Q(\pi) = \frac{n_A}{n\,\tau_s}\Big[1 - \Phi\Big(\frac{z - m_U}{\sigma_U\sqrt{\rho_n}}\Big)\Big].$$

Year 0 of the illustration: $n = 3$, $z \approx 0.21$, $P_\pi(\text{accept}) = 0.15$, and $Q = 0.15/(3 \times 0.05) = 1$ plan a year $= K$.

Napoleon's trade-off: how selective he can be

With n signals per plan, each plan takes $n\tau_s$. So n_A Napoleons examine

$$R_n = \frac{n_A}{n\,\tau_s} \quad \text{plans per unit of time.}$$

Capacity lets K of them through, so Napoleon accepts the top share $q_n = K/R_n = Kn\tau_s/n_A$. He does so with a **cutoff** on his estimate, never accepting a plan worth less than nothing:

$$\text{accept} \iff \mu_n \geq z_n, \qquad z_n = \max\left\{0,\ m_U + \sigma_U\sqrt{\rho_n}\,\Phi^{-1}(1-q_n)\right\}.$$

> More signals per plan: fewer plans examined, so Napoleon must be less selective.

Napoleon's trade-off: how good his picks are

Napoleon accepts the plans with the highest estimates μ_n. Since $\mathbb{E}[U \mid \mu_n] = \mu_n$,

$$\mathbb{E}[U \mid \text{accept}] = \mathbb{E}\big[\mu_n \mid \mu_n \text{ in its top share } q_n\big].$$

Under the Gaussian-plans conditions, $\mu_n = m_U + \sigma_U\sqrt{\rho_n}\,X$ with $X \sim N(0,1)$. The top share q of X has mean $S(q) = \phi\big(\Phi^{-1}(1-q)\big)/q$, so

$$\mathbb{E}[U \mid \text{accept}] = m_U + \sigma_U\sqrt{\rho_n}\; S(q_n).$$

$S(q)$ is the inverse Mills ratio: $S(0.05) = 2.06$, $S(0.15) = 1.55$, $S(0.5) = 0.80$, $S(1) = 0$.

Napoleon's trade-off: output

Output is the accepted flow times the value of each accepted plan:

$$y(n) = K\Big(m_U + \sigma_U \sqrt{\rho_n}\, S(q_n)\Big).$$

- K: plans carried out per unit of time;
- m_U: the value of an average plan;
- σ_U: how much plans differ, i.e. what there is to gain from choosing well;
- $\sqrt{\rho_n}$: how much of that difference Napoleon can see after n signals;
- $S(q_n)$: how far above average, in standard deviations, the plans he picks are.

Assumes capacity binds and the cutoff is positive; otherwise Napoleon accepts fewer than K plans.

Proof of Lemma 1

Write $\varepsilon = \varepsilon_B$. Because $|\hat{W} - W| \le \varepsilon$ everywhere and $\hat{a}$ maximizes $\hat{W}$,

$$W(\hat{a};\xi) \;\ge\; \hat{W}(\hat{a};\xi) - \varepsilon \;\ge\; \hat{W}\big(a^*(\xi);\xi\big) - \varepsilon \;\ge\; W\big(a^*(\xi);\xi\big) - 2\varepsilon.$$

So $\kappa_B = |\upsilon(\xi) - W(\hat{a};\xi)| \le 2\varepsilon$. The rate $\varepsilon \le C_s k^{-s/d}$ is a Jackson-type bound for polynomial and spline sieves. □

Proof of Lemma 2

Let $X = Z - \mathbb{E}[Z]$. For any $\theta > 0$, Jensen's inequality gives

$$e^{\theta \mathbb{E}[X \mid A]} \;\le\; \mathbb{E}\big[e^{\theta X} \mid A\big] \;\le\; \frac{\mathbb{E}\big[e^{\theta X}\big]}{q}.$$

Take logs and use $\ln \mathbb{E}[e^{\theta X}] \le \theta^2 \sigma^2 / 2$:

$$\mathbb{E}[X \mid A] \;\le\; \frac{\theta \sigma^2}{2} + \frac{\ln(1/q)}{\theta}.$$

Choose $\theta = \sqrt{2 \ln(1/q)}/\sigma$. $\square$

Lemma A1 (screening bound)

Statement

For any policy, output is bounded by the rate of screening: $\mathcal{Y} \le \bar{R}\,\mathbb{E}[U^+]$*, where* $\bar{R} = n_A/\tau_s$.

Proof. Each plan needs at least one signal, so plans are examined at rate $R \le \bar{R}$. With $A = \{\text{accept}\}$,

$$\mathcal{Y} = Q\,\mathbb{E}[U \mid A] = R\,\mathbb{E}[U\mathbf{1}_A] \;\le\; \bar{R}\,\mathbb{E}[U^+]. \qquad \square$$

Proof of Proposition 1

- Whatever Berthier does, the realized action lies in $\mathcal{R}$, so $Y \le \bar{U}(\xi)$ and hence $U_t \le \bar{U}$.
- By Lemma A1, $\mathcal{Y}_t \le \bar{R}\,\mathbb{E}[U_t^+] \le \bar{R}\,\mathbb{E}[\bar{U}^+]$.
- $\bar{U} \le v$, so $\mathbb{E}[\bar{U}^+] \le \mathbb{E}[v^+] < \infty$.
- A bounded sequence has $\limsup_t \frac{1}{t} \ln \mathcal{Y}_t \le 0$. □

Proof of Proposition 2

Accepted plans flow at $Q = qR \le K$, with $R \le \bar{R}$. Since $U \le \bar{U}$, Lemma 2 with $Z = \bar{U}$ gives

$$\mathcal{Y} = Q\,\mathbb{E}[U \mid A] \;\le\; Q\,\mathbb{E}[\bar{U} \mid A] \;\le\; Q\,\bar{m}^{+} + \sigma f(Q,R), \qquad f(Q,R) = Q\sqrt{2\ln(R/Q)}.$$

- f is increasing in R, and in Q when $Q \le R/\sqrt{e}$.
- With $R \le \bar{R}$ and $Q \le K \le \bar{R}/\sqrt{e}$: $f \le K\sqrt{2\ln(\bar{R}/K)}$. □

Proof of Proposition 3

- Proposition 2 holds at each date t, for any Berthier.
- Its bound rises with K (for $K \leq \bar{R}_t/\sqrt{e}$), so use $K_t \leq \mu_L$.
- Substitute $\bar{R}_t \leq \bar{R}_0 e^{gt}$.
- A bound of order $\sqrt{t}$ implies $\limsup_t \frac{1}{t} \ln y_t \leq 0$. □

Proof of the heavy-tail result

- No policy beats perfect sorting: $\mathbb{E}[U \mid \text{accept}] \le M(q)$, the mean of the top share q of U.
- For a Pareto tail, $M(q) = \frac{\alpha}{\alpha-1} x_m q^{-1/\alpha}$.
- So $\mathcal{Y}_t \le \mu_L \, M(\mu_L / \bar{R}_t)$, which grows at rate at most g/α.
- With perfectly informative signals Napoleon sorts perfectly, so $\mathcal{Y}_t = K \, M(K/\bar{R}_t)$, which grows at exactly g/α. □

Proof of Proposition 4

- The realized action is some $r \in \mathcal{R}$, so $r \le \bar{r}$; the bound on Y then also holds for its expectation U.
- For fixed r, $\beta a^* - \frac{H}{2}(r - a^*)^2$ is largest at $a^* = r + \beta/H$, so

$$Y \;\le\; \beta r + \frac{\beta^2}{2H} \;\le\; \beta \bar{r} + \frac{\beta^2}{2H}.$$

- At most μ_L plans are carried out per unit of time. □

Proof of Proposition 5

- The realized action lies in $\mathcal{R} \subseteq \mathcal{A}(\xi)$, so $U_t \leq \upsilon$ whatever Berthier and Lannes do.
- By Lemma A1, $\mathcal{Y}_t \leq \frac{n_A}{\tau_s}\mathbb{E}[U_t^+] \leq \frac{n_A}{\tau_s}\mathbb{E}[\upsilon^+]$.
- Every policy has $Q(\pi) \leq n_A/\tau_s$. If $K_t = \min\{\mu_{B,t}, \mu_{L,t}\} \geq n_A/\tau_s$, then $Q(\pi) \leq K_t$ for every policy: the constraint is slack. □

Proof of Proposition 6

- In research $U = \upsilon - \kappa_B \leq \upsilon$, because $\kappa_B \geq 0$.
- By Lemma A1, $\mathcal{Y}_t \leq \frac{n_A}{\tau_s}\mathbb{E}[U_t^+] \leq \frac{n_A}{\tau_s}\mathbb{E}[\upsilon^+]$, whatever Berthier does.
- Each plan needs at least one signal, so every policy has $Q(\pi) \leq R(\pi) \leq n_A/\tau_s$. If $\mu_{B,t} \geq n_A/\tau_s$, then $Q(\pi) \leq K_t$ for every policy: the constraint is slack. □

Proof of Proposition 7

Write $\bar{R}_t = n_A/\tau_{s,t}$ and $K_t = \mu_{B,t}$; research means $\bar{U} = v \sim N(m, \sigma^2)$.

- *Upper bound.* Lemma A1 gives $\mathcal{Y}_t \le \bar{R}_t \mathbb{E}[v^+]$, growing at rate g. If $g_B < g$, then $\bar{R}_t \ge \sqrt{e} K_t$ eventually and Proposition 2 gives $\mathcal{Y}_t \le K_t\big(m^+ + \sigma\sqrt{2\ln(\bar{R}_t/K_t)}\big)$, growing at rate g_B. So $g_{\mathcal{Y}} \le \min\{g_B, g\}$.
- *Lower bound.* Take one signal and accept iff $\mu_1 \ge \max\{c, z_t\}$, for a fixed $c > 0$ and the capacity cutoff z_t. Accepted plans have $\mathbb{E}[U \mid \text{accept}] \ge c$, and $P(\mu_1 \ge c) \ge p > 0$ for all t, since AI only raises m_U and signals do not get noisier. So $\mathcal{Y}_t \ge c \min\{\bar{R}_t p,\ K_t\}$, growing at rate $\min\{g, g_B\}$. □